

Metamodel definitions

DCL

FamilyStatus

Property

Income

Applicant

has_ch

owns_p

earns

num_children

property_area

monthly_income

C4: $\text{area} \geq f(\text{ch})$

C2

C2

C2

FamilyStatusCred

CS_App1

claim $\rightarrow$ numch1

PropertyCred

CS_App2

claim $\rightarrow$ area1

IncomeCred

CS_App3

claim $\rightarrow$ inc1

C3

C1: entity alignment
C3

C3

(1) CHECK

UNSAT under
 $\text{eIDAS} \wedge \text{GDPR} \wedge \text{W3C}$

Scenario instance

(housing subsidy)

CSL

(2) IDENTIFY

governance_conflict
cross_cred_gap

Governance constraints

C5: eIDAS

C6: GDPR

C7: W3C

FSL

SD-JWT-VC

eIDAS

SD-JWT-VC

eIDAS

C9: cross-cred gap
no format supports
cross-credential ZKP

C5

SD-JWT-VC

eIDAS: ok

pred: NO

W3C: ok

C6

vs.

C7

AnonCreds

eIDAS: NO

pred: ok

W3C: NO

(3) GENERATE

SAT when eIDAS
removed $\rightarrow$ AnonCreds

[Refinery SVG: concrete.svg]
generated valid instance

$C5 \wedge C6 \wedge C7 \Rightarrow \text{UNSAT}$

INPUT

MODEL

OUTPUT